C Programming - Day 9 Practical Report

Question 1

Aim: Develop a menu-driven C program that exchanges two values by using a temporary variable as well as by using arithmetic operations without an extra variable.

Algorithm

1. Begin execution.

2. Show a menu containing three choices: swap with a temporary variable, swap without a temporary variable, or exit.

3. Accept the menu selection from the user.

4. For option 1, read two values, show the original values, interchange them with a temporary variable, and print the swapped values.

5. For option 2, read two values and exchange them through addition and subtraction without declaring another variable; then display the result.

6. When option 3 is selected, end the menu loop.

7. For any unsupported option, print an appropriate invalid-selection message.

8. Continue displaying the menu until Exit is selected.

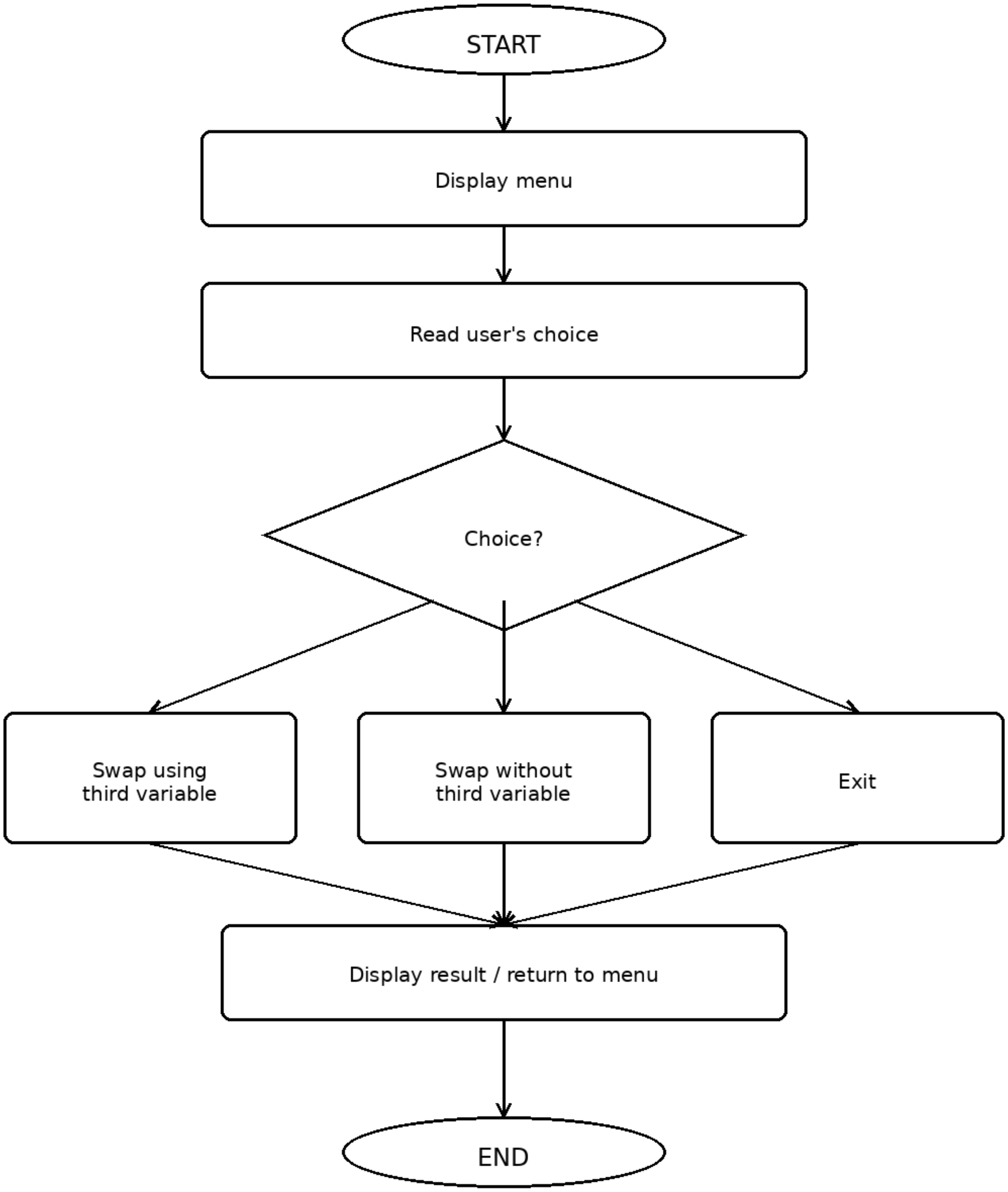
9. End the program.

Pseudocode

```
BEGIN
REPEAT
    DISPLAY swap menu
    READ choice
    IF choice = 1 THEN
        INPUT a, b
        DISPLAY values before swapping
        temp ← a
        a ← b
        b ← temp
        DISPLAY values after swapping
    ELSE IF choice = 2 THEN
        INPUT a, b
        DISPLAY values before swapping
        a ← a + b
        b ← a - b
        a ← a - b
        DISPLAY values after swapping
```

```
    ELSE IF choice is not 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

Flowchart


START
Display menu
Read user's choice
Choice?
Swap using
third variable
Swap without
third variable
Exit
Display result / return to menu
END

C Program

```c
#include <stdio.h>

int main() {
    int choice;
    float a, b, temp;

    do {
        printf("\n--- SWAP TWO NUMBERS ---\n");
        printf("1. Swap using third variable\n");
        printf("2. Swap without third variable\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                printf("Before swapping: a = %.2f, b = %.2f\n", a, b);
                temp = a;
                a = b;
                b = temp;
                printf("After swapping: a = %.2f, b = %.2f\n", a, b);
                break;

            case 2:
                printf("Enter two numbers: ");
                scanf("%f %f", &a, &b);
                printf("Before swapping: a = %.2f, b = %.2f\n", a, b);
                a = a + b;
                b = a - b;
                a = a - b;
                printf("After swapping: a = %.2f, b = %.2f\n", a, b);
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);
```

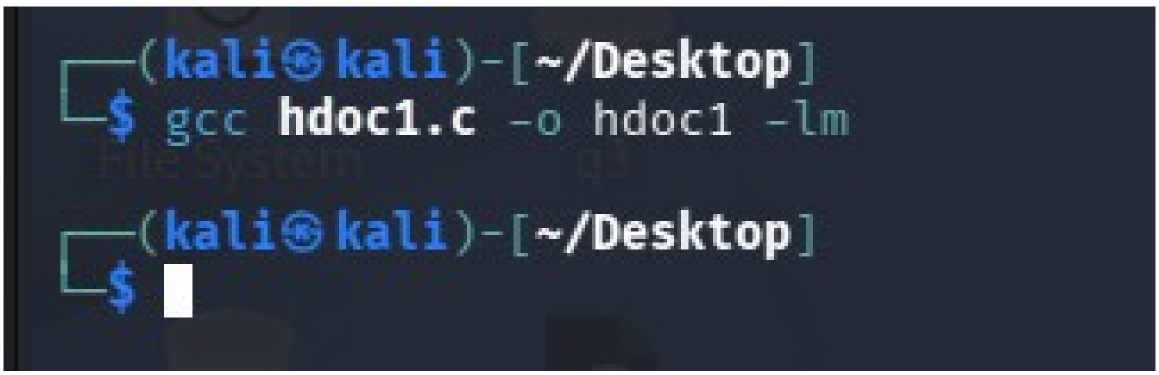
```
    return 0;
}
```

Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
TC1	Choice 1; a=10, b=20	a=20, b=10	a=20, b=10	PASS
TC2	Choice 2; a=15, b=30	a=30, b=15	a=30, b=15	PASS
TC3	Choice 9	Invalid choice message	Invalid choice message	PASS

Program Output Screenshots

Screenshot 1



Screenshot 2

Screenshot 3

```
═══════════ MENU ═══════════
1. Swap using third variable
2. Swap without third variable
3. Exit
Enter your choice: 2

Enter two numbers: 15
30

Before swapping:
a = 15
b = 30

After swapping:
a = 30
b = 15
```

Screenshot 4

```
═══════════ MENU ═══════════
1. Swap using third variable
2. Swap without third variable
3. Exit
Enter your choice: 3

Exiting Program...
```

Question 2

Aim: Create a menu-based C program to compute either the sum of the first n natural numbers or the sum of their squares.

Algorithm

1. Begin execution.

2. Present choices for calculating the natural-number sum, the sum of squares, or exiting the program.

3. Accept the menu selection from the user.

4. For the first option, read n, ensure it is positive, and evaluate n(n + 1)/2.

5. For the second option, validate n and compute n(n + 1)(2n + 1)/6.

6. Print the computed value on the screen.

7. If choice is 3, terminate the program.

8. Otherwise, display an invalid-choice message.

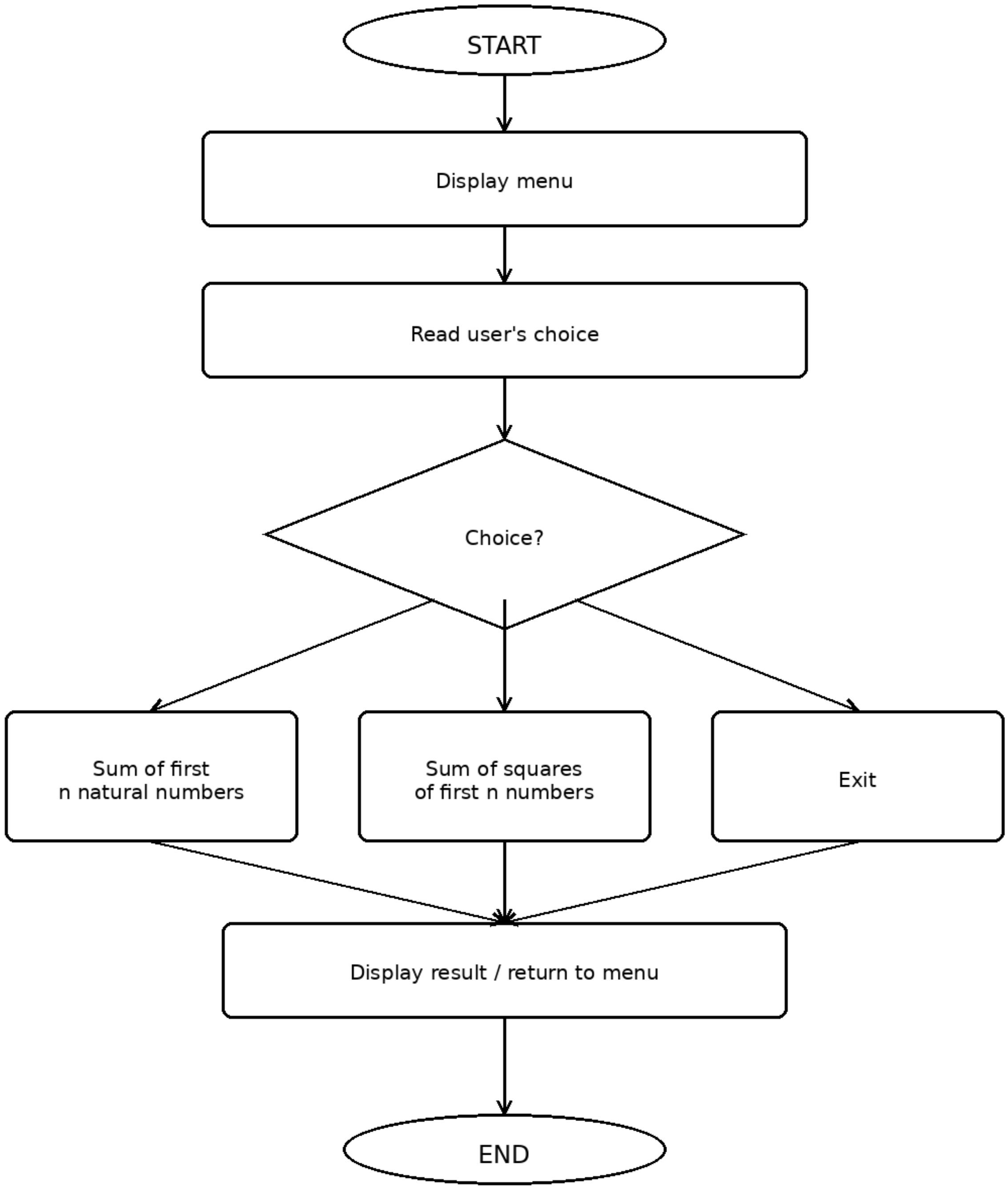
9. Repeat the menu until the user chooses Exit.

10. Stop.

Pseudocode

```
BEGIN
REPEAT
    DISPLAY calculation menu
    READ choice
    IF choice = 1 THEN
        INPUT n
        IF n > 0 THEN
            sum ← n × (n + 1) / 2
            DISPLAY sum
        END IF
    ELSE IF choice = 2 THEN
        INPUT n
        IF n > 0 THEN
            squareSum ← n × (n + 1) × (2n + 1) / 6
            DISPLAY squareSum
        END IF
    ELSE IF choice is not 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

Flowchart



C Program

```c
#include <stdio.h>

int main() {
    int choice, n, i;
    long long sum = 0, squareSum = 0;

    do {
        printf("\n--- SUM CALCULATIONS ---\n");
        printf("1. Sum of first n natural numbers\n");
        printf("2. Sum of squares of first n natural numbers\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter n: ");
                scanf("%d", &n);
                if (n > 0) {
                    sum = (long long)n * (n + 1) / 2;
                    printf("Sum of first %d natural numbers = %lld\n", n, sum);
                } else {
                    printf("Please enter a positive natural number.\n");
                }
                break;

            case 2:
                printf("Enter n: ");
                scanf("%d", &n);
                if (n > 0) {
                    squareSum = (long long)n * (n + 1) * (2 * n + 1) / 6;
                    printf("Sum of squares of first %d natural numbers = %lld\n",
                           n, squareSum);
                } else {
                    printf("Please enter a positive natural number.\n");
                }
                break;

            case 3:
                printf("Exiting program...\n");
                break;
```

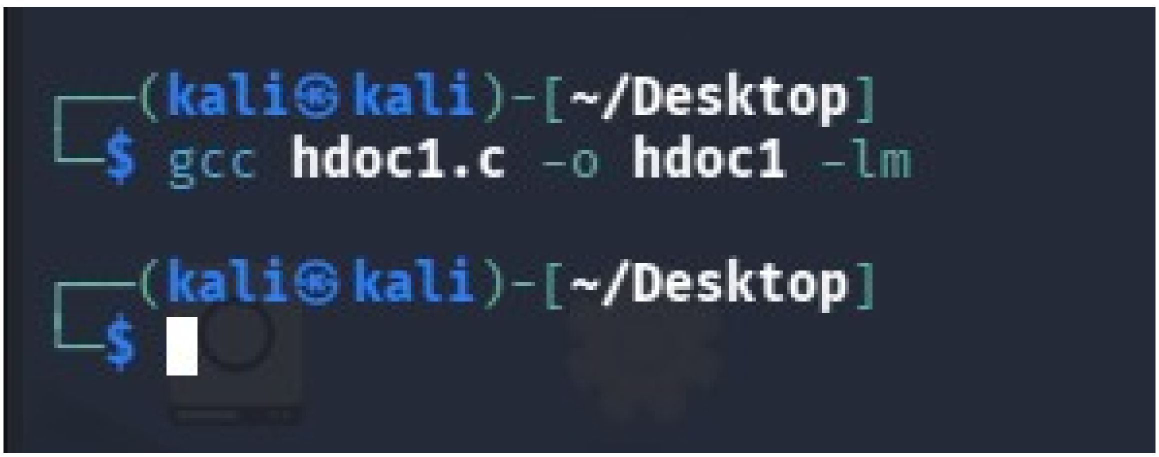
```
            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);


    return 0;
}
```

Test Case Table

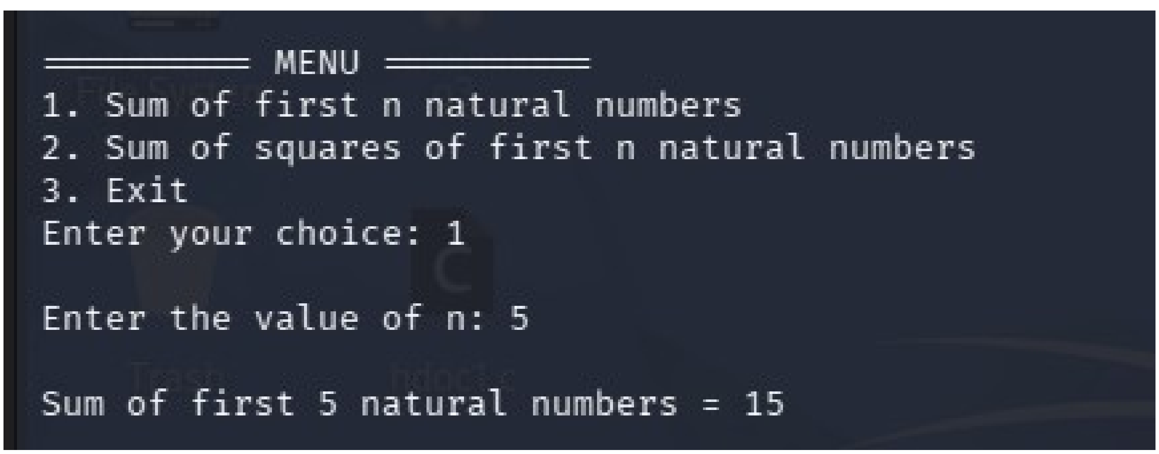
Test Case	Input	Expected Output	Actual Output	Result
TC1	Choice 1; n=5	15	15	PASS
TC2	Choice 2; n=5	55	55	PASS
TC3	Choice 1; n=-2	Positive-number message	Positive-number message	PASS

Program Output Screenshots

Screenshot 1



Screenshot 2



Screenshot 3

```
════════════ MENU ════════════
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2

Enter the value of n: 5

Sum of squares of first 5 natural numbers = 55
```

Screenshot 4

```
════════════ MENU ════════════
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 3

Exiting Program...
```

Question 3

Aim: Design a C program to perform common interest calculations from principal, rate and time, including simple interest, compound amount, compound interest and payable amount.

Algorithm

1. Begin execution.

2. Display a menu for simple-interest calculation, compound calculations, and program exit.

3. Accept the menu selection from the user.

4. For simple interest, accept principal, rate and time, evaluate SI = (P x R x T)/100, and add it to the principal for the total.

5. For compound calculations, read P, R and T, determine A = P(1 + R/100)^T, and obtain CI by subtracting P from A.

6. Display the relevant interest, amount, and payable value.

7. If choice is 3, terminate the program.

8. Otherwise, display an invalid-choice message.

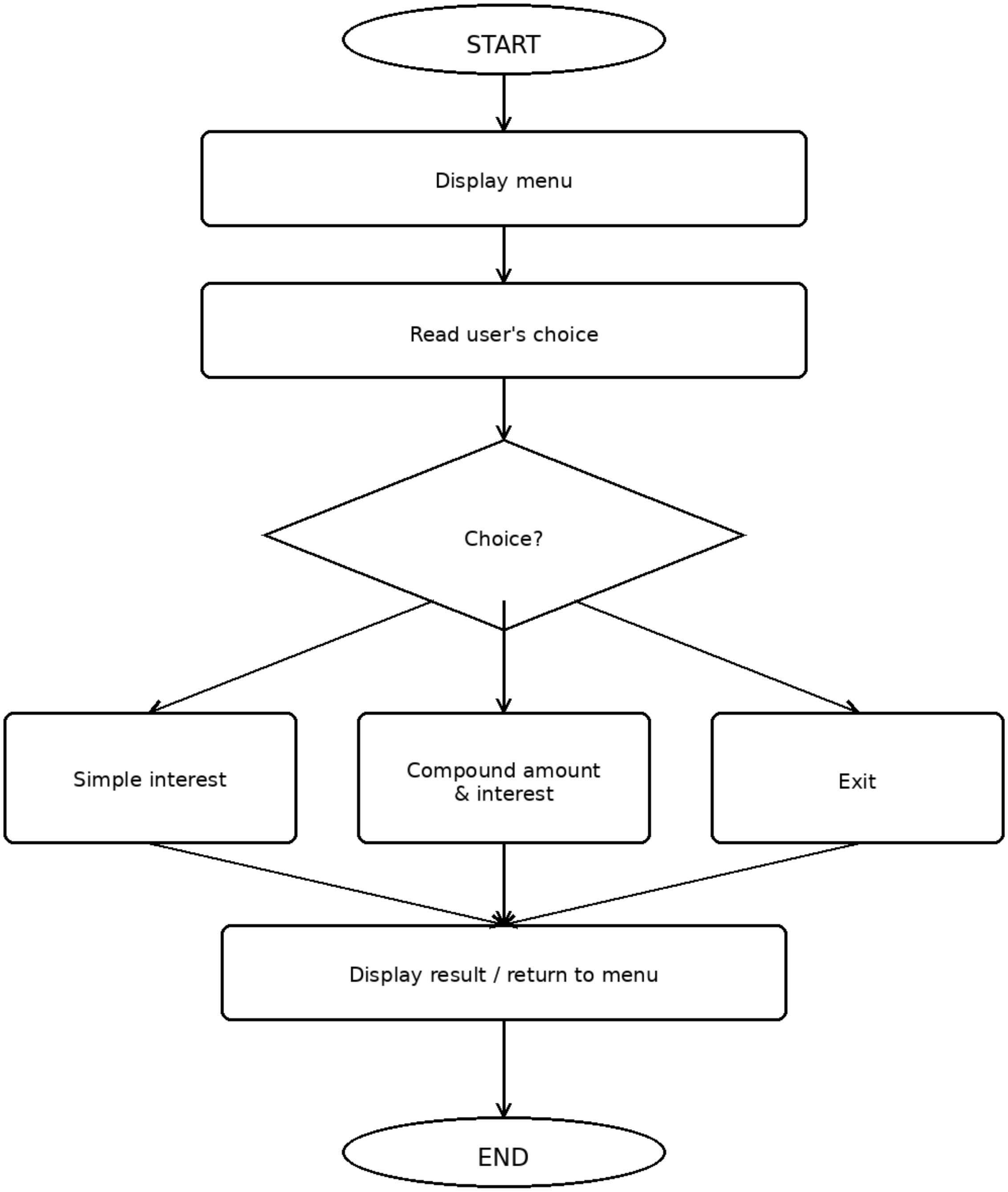
9. Repeat the menu until the user chooses Exit.

10. Stop.

Pseudocode

```
BEGIN
REPEAT
    DISPLAY interest menu
    READ choice
    IF choice = 1 THEN
        INPUT P, R, T
        SI ← (P × R × T) / 100
        Total ← P + SI
        DISPLAY SI and Total
    ELSE IF choice = 2 THEN
        INPUT P, R, T
        A ← P × (1 + R/100)^T
        CI ← A - P
        DISPLAY A, CI and Total Payable Amount
    ELSE IF choice is not 3 THEN
        DISPLAY "Invalid choice"
    END IF
UNTIL choice = 3
END
```

Flowchart


START
Display menu
Read user's choice
Choice?
Simple interest
Compound amount
& interest
Exit
Display result / return to menu
END

C Program

```c
#include <stdio.h>
#include <math.h>

int main() {
    int choice;
    double principal, rate, time, simpleInterest;
    double compoundAmount, compoundInterest;

    do {
        printf("\n--- INTEREST CALCULATIONS ---\n");
        printf("1. Calculate Simple Interest\n");
        printf("2. Calculate Compound Amount and Compound Interest\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter principal, rate and time: ");
                scanf("%lf %lf %lf", &principal, &rate, &time);

                simpleInterest = (principal * rate * time) / 100;
                printf("Simple Interest = %.2lf\n", simpleInterest);
                printf("Total Payable Amount = %.2lf\n",
                       principal + simpleInterest);
                break;

            case 2:
                printf("Enter principal, rate and time: ");
                scanf("%lf %lf %lf", &principal, &rate, &time);

                compoundAmount = principal *
                    pow((1 + rate / 100), time);
                compoundInterest = compoundAmount - principal;

                printf("Compound Amount = %.2lf\n", compoundAmount);
                printf("Compound Interest = %.2lf\n", compoundInterest);
                printf("Total Payable Amount = %.2lf\n", compoundAmount);
                break;

            case 3:
                printf("Exiting program...\n");
```

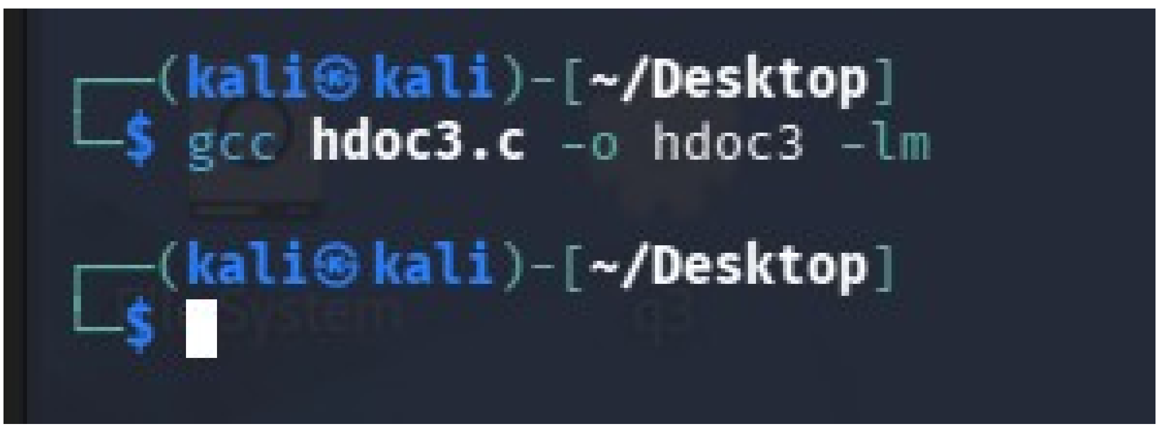
```
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Test Case Table

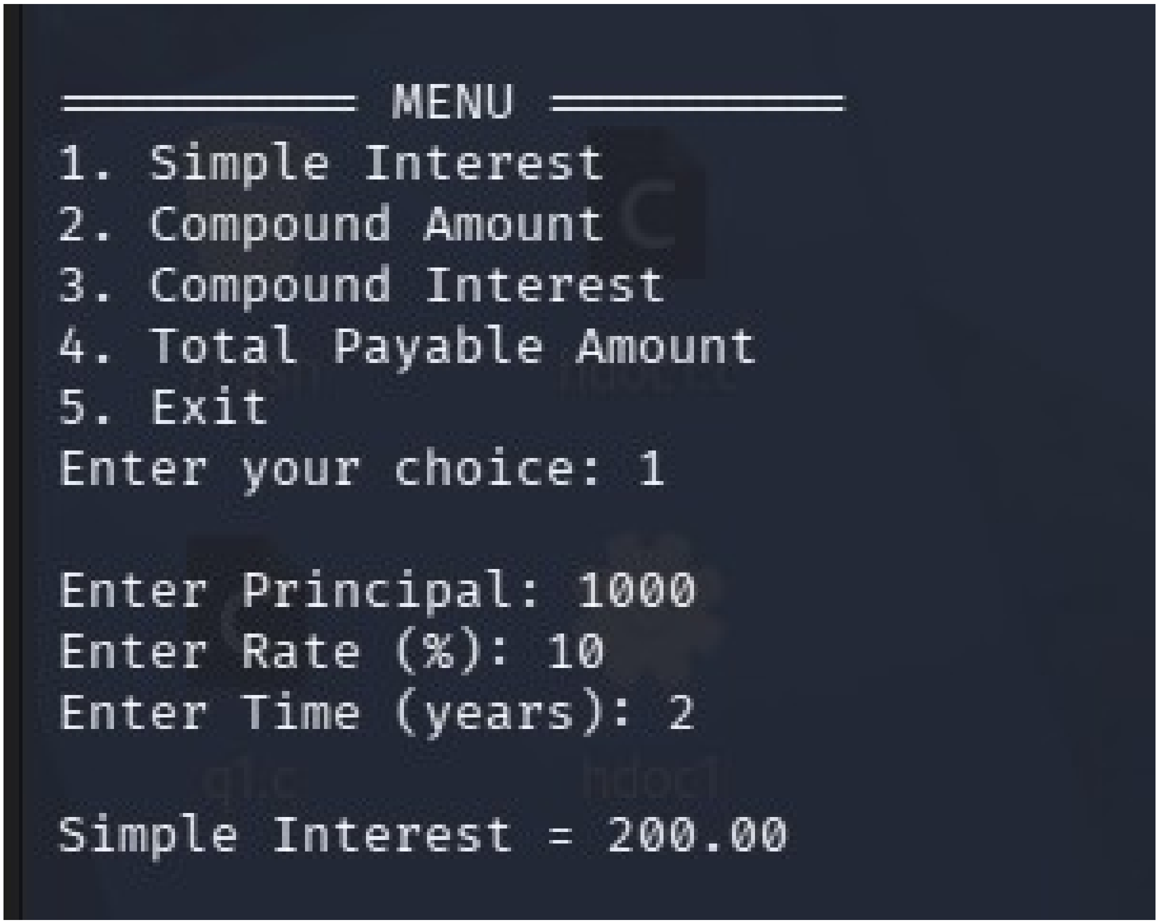
Test Case	Input	Expected Output	Actual Output	Result
TC1	Choice 1; P=10000, R=10, T=2	SI=2000; Total=12000	SI=2000; Total=12000	PASS
TC2	Choice 2; P=10000, R=10, T=2	A=12100; CI=2100	A=12100; CI=2100	PASS
TC3	Choice 9	Invalid choice message	Invalid choice message	PASS

Program Output Screenshots

Screenshot 1



Screenshot 2



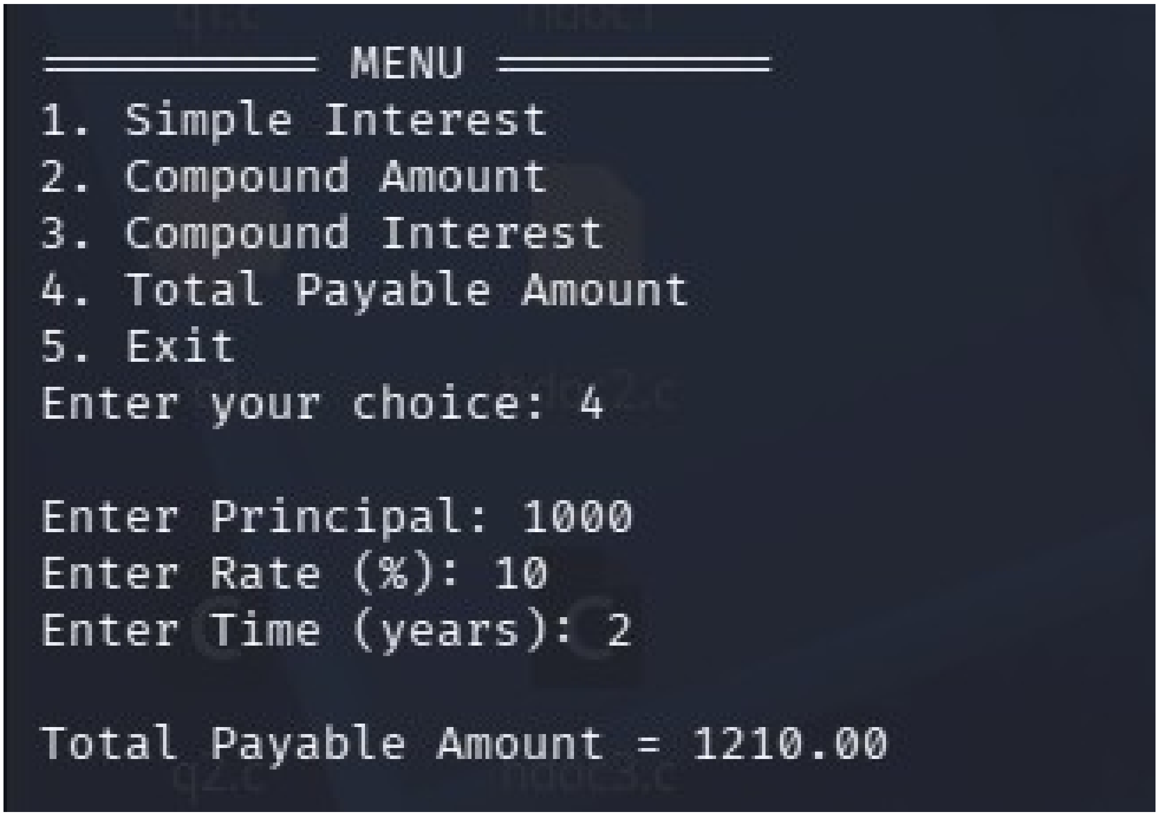
Screenshot 3

```
═══════════ MENU ═══════════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 2

Enter Principal: 1000
Enter Rate (%): 10
Enter Time (years): 2

Compound Amount = 1210.00
```

Screenshot 4

```
═══════════ MENU ═══════════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 3

Enter Principal: 1000
Enter Rate (%): 10
Enter Time (years): 2

Compound Interest = 210.00
```

Screenshot 5



```
════════════ MENU ════════════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 4

Enter Principal: 1000
Enter Rate (%): 10
Enter Time (years): 2

Total Payable Amount = 1210.00
```

Screenshot 6

```
════════════ MENU ════════════
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 5

Exiting Program...

┌──(kali㉿kali)-[~/Desktop]
└─$
```